

Multi-biome assessment of tree water use resilience to drought

Brenda V. Fatecha^{a,b,*}, Jordi Martínez-Vilalta^{a,b}, Maurizio Mencuccini^{a,c},
Rafael Poyatos^{a,b}

^a CREAF, Ecological and Forestry Applications Research Centre, E08193 Bellaterra (Cerdanyola del Vallès), Catalonia, Spain

^b Department of Animal Biology, Plant Biology and Ecology, Autonomous University of Barcelona (UAB), E08193 Bellaterra (Cerdanyola del Vallès), Catalonia, Spain

^c Institució Catalana de Recerca i Estudis Avançats (ICREA), Barcelona, Catalonia E08010, Spain

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ABSTRACT

Drought frequency and severity are increasing in many regions, impacting forest functioning and dynamics, with effects that can result in widespread tree death and ecosystem shifts. These events have been linked to soil and atmospheric hydrological changes, but the causes of different responses in tree water-use are poorly understood at the global scale. We used data from the SAPFLUXNET global database to assess water-use resilience to drought of > 900 trees of 96 different species in 89 plots distributed across the globe. We detected decreases in water-use resilience when soil water availability was lower before, during or after the drought. Atmospheric drought enhanced resilience during the pre-drought period, but diminished it post-drought. Resilience was also favoured by longer recovery periods since the last, previous drought, and in larger diameter trees. Hydraulic and water relations traits also played a role in determining differences in tree water-use resilience across species. Species that were more vulnerable to stem xylem embolism (less negative P50) or had a more negative turgor loss point exhibited higher water-use resilience (and resistance). Our findings help to better understand the drivers of water use resilience and therefore contribute to our understanding of how trees respond to severe droughts at multiple spatio-temporal scales.

1. Introduction

Drought severity (intensity and duration) and frequency are increasing in many regions, impacting forest functioning (carbon cycling, hydrologic regulation) and dynamics (IPCC, 2023). These effects can be seen in the widespread tree death and ecosystem shifts that are occurring globally (Allen et al., 2015; Brodribb et al., 2020; Hartmann et al., 2022; McDowell et al., 2020). Although the determinants of tree mortality have been assessed in previous studies (Hammond et al., 2022; McDowell et al., 2022; Sanchez-Martinez et al., 2023), the effects of droughts of varying duration and intensity, which do not necessarily cause immediate mortality but determine long-term forest responses, including future mortality risk, are not clearly understood (DeSoto et al., 2020; Tai et al., 2023; Wu et al., 2022). Trees undergoing a non-lethal drought may experience severe hydraulic damage and reductions in growth and leaf area, which may have long-term consequences on their functioning, including carbon and water fluxes (Vilonen et al., 2022). Recurrent droughts may further impair tree functioning and increase forest vulnerability (Anderegg et al., 2020). Therefore, quantifying

drought resilience and understanding its mechanisms is essential to predict forest functioning in the face of increasing climate-driven disturbances.

Research on the capacity of trees and forests to absorb or withstand the effects of disturbances is usually approached using the resilience concept (Lecina-Diaz et al., 2024; Nikinmaa et al., 2020; Schwarz et al., 2020; Vilonen et al., 2022). Resilience in ecology has frequently been quantified from time series, by using metrics that measure disturbance impact and recovery relative to a baseline (Ingrisch and Bahn, 2018). Studies assessing the resilience of forest ecosystems to drought have characteristically compared the value of an ecological variable after the disturbance with the value of the same variable in the absence of disturbance (e.g., before it occurred). In these approaches, two components of resilience are typically considered: resistance, which refers to the capacity to minimise the immediate impact of the disturbance; and recovery, which refers to the capacity to bounce back after the disturbance. Studies of tree and forest resilience have largely focused on interannual patterns using radial growth from tree ring series (DeSoto et al., 2020; Lloret et al., 2011), although other variables have been

* Corresponding author.

E-mail address: b.fatecha@creaf.uab.cat (B.V. Fatecha).

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used, including vegetation indices obtained from remote sensing (Forzieri et al., 2022). While studies utilising tree-ring analysis, remote sensing, and ecosystem-level assessments have provided valuable insights into the impact of climate (Schwalm et al., 2017), drought characteristics and recurrence (Anderegg et al., 2020; Jiao et al., 2021), and plant traits (Anderegg et al., 2018; Hu et al., 2022) on drought forest resilience, these approaches primarily examine the cumulative or net long-term effect on vegetation growth or greenness. A more physiological perspective is needed to understand the mechanisms that determine tree and forest responses at shorter time scales (Ruehr et al., 2019), and how they constrain longer-term dynamics (DeSoto et al., 2020; Kannenberg et al., 2020). Specifically, we propose here that a more detailed understanding of tree water use resilience in the short-term (weeks to months) is likely to provide key insights into the factors that determine the functional response of forests to more severe and frequent droughts.

The regulation of water transport and use at the tree level is crucial to understand how forests respond to drought. This is because water-related (hydraulic) traits are intimately involved in the process of drought-induced tree mortality (Brodribb et al., 2020; McDowell et al., 2022) and also determine water use and drought response strategies (Flo et al., 2021; Kannenberg et al., 2022). Although ecosystem-level approaches using eddy covariance data provide important insights into cross-biome patterns of forest functional resilience to drought (Shekhar et al., 2023; Xu et al., 2023), one current limitation of these approaches is the challenge to precisely separate tree transpiration from water evaporation from soil and leaves, which hinders a deeper understanding of the underlying mechanisms. Focusing on tree-level water use offers a complementary approach, as it allows a direct link to mechanisms and plant functional traits (Flo et al., 2021; Myers-Smith et al., 2019; Smith, 2011), and provides a natural way to characterise the response of the different species coexisting in mixed forests. In addition, the availability of cross-biome tree water use data from sap flow measurements in the SAPFLUXNET database (Poyatos et al., 2021) opens the possibility of assessing the global drivers of tree water use resilience.

When studying resilience in water use at the tree level, several factors need to be accounted for. Firstly, drought characteristics (i.e., its duration and intensity) and environmental conditions before, during and after the drought can significantly influence tree resilience. Less severe droughts and higher soil water availability prior to and following the drought tend to facilitate the recovery, which may also be enhanced by lower post-drought vapour-pressure deficit (VPD) (Liu et al., 2018; Yao et al., 2023), that is, the difference between the actual amount of moisture in the air and its saturation point at a given temperature. VPD is a central variable because it determines the atmospheric water demand for plants, directly impacting their growth and health status (Novick et al., 2024). In addition, longer droughts could lead to more persistent physiological declines, even relative to more intense but shorter droughts (Jiao et al., 2021). On the other hand, recurrent droughts are known to trigger legacy effects that hamper the recovery of vegetation productivity after drought (Jiao et al., 2021; Schwalm et al., 2017; Zhang et al., 2021). Similar effects are likely to occur for the recovery of water fluxes, although we are not aware of any study focusing on those.

Secondly, site properties such as climate, soil, and vegetation structure (including biotic interactions; Haberstroh and Werner, 2022) are likely to modulate drought resilience. In general, less productive sites are expected to need longer recovery times than more fertile ones, at least in terms of gross primary productivity and water use efficiency (Yu et al., 2017). Finally, species attributes, including tree size, will likely determine the observed responses (Ruehr et al., 2019). Current evidence indicates that in certain woody plant species, hydraulic impairment (damage to the water transport system) constrains their ability to recover gas exchange following a drought (Brodribb and Cochard, 2009; Kannenberg et al., 2019; Skelton et al., 2017), suggesting that species with a more resistant hydraulic system (i.e., those less likely to suffer from embolism) may be more resilient and better able to recover from

such events. However, some studies have found faster recovery of leaf water potential and net photosynthetic rates in more hydraulically vulnerable species (Yin and Bauerle, 2017) while others have found no clear difference in gas exchange recovery among functionally diverse species (Skelton et al., 2023). Stomatal control is another key aspect determining the impact of drought on water use and its recovery. Assuming other factors are constant, species with stricter stomatal control are expected to reduce water fluxes more strongly during drought (i.e., lower resistance), and may also need more time to return to pre-drought fluxes after the drought ends (i.e., slower recovery) (Hesse et al., 2023; Ulrich and Grossiord, 2023). Clearly, the drivers of water use resilience and the relevant plant traits are less understood than those influencing tree water use strategies in response to VPD and soil water availability (Flo et al., 2021, 2022). The effect of tree size is particularly important in this context, as taller trees are usually considered to be more vulnerable to drought (and potentially less able to recover functionality) than shorter ones (Bennett et al., 2015; Fernández-de-Uña et al., 2023; McGregor et al., 2021; Stovall et al., 2019). Identifying the traits driving water use resilience will likely improve our understanding and capacity to predict the impact of drought on forest mortality and on long-term ecosystem function.

Here, we assess the resilience to drought of tree-level water use at daily to seasonal scales, taking advantage of the SAPFLUXNET global database of sap flow measurements (Poyatos et al., 2021). Our main objective is to identify the main biotic and abiotic drivers that determine the observed patterns of tree water use resilience to drought, including climate, soil, stand- and tree-level attributes, including species traits and drought characteristics. Our specific hypotheses are:

1. Drought characteristics will affect tree water-use resilience, with higher soil water availability and milder atmospheric drought conditions previously to, during and/or after individual drought events promoting resilience. In contrast, recurrent droughts will reduce water use resilience.
2. Climate, stand and soil characteristics will also influence tree water use resilience. We expect that sites that are more humid or fertile will be characterised by higher tree resilience to drought than more arid or poorer sites (after accounting for confounding effects).
3. Species-level hydraulic and water relations traits will predict water-use resilience, with traits associated with less strict stomatal regulation or higher hydraulic resistance conferring higher resilience.

2. Methodology

To address our main objective, we followed a three-step process: first we identified edaphic droughts in the SAPFLUXNET database (Poyatos et al., 2021), then we calculated tree-level water use resilience metrics in response to such droughts and, finally, we analysed how resilience (and its components) varied as a function of drought characteristics, site factors and tree-level variables.

2.1. The SAPFLUXNET dataset

The SAPFLUXNET database (Poyatos et al., 2021) is a comprehensive collection of sap flow measurements from diverse ecosystems worldwide. It integrates sap flow data collected through different methodologies, including thermal dissipation probes, heat balance, heat pulse methods, and other established techniques. Consequently, efforts have been made to ensure data quality and consistency across studies, including standardising data formats, units, and metadata descriptions to facilitate interoperability and comparability. We also note that the use of resilience ratios (see later) largely helps to minimise the impact of combining different sap flow techniques in the same analysis. We downloaded the version 0.1.5 of the database (Poyatos et al., 2020), which provides data at hourly or even higher temporal resolution, and aggregated sub-daily sap flow per sapwood area and in situ

meteorological data to daily means. We considered using midday values, but due to their strong correlation with daily averages ($R^2 = 0.93$), and the potential for midday values to overemphasize VPD-driven changes, we opted to use daily means. The database includes sap flow data for 202 datasets and 174 species (2714 trees in total) covering all terrestrial biomes (with woodland/shrubland plus temperate forest biomes adding up to 80 % of the datasets). The oldest datasets go back to 1995 while the most recent data reaches up to 2018. Several multi-year datasets are present, with 50 % of the datasets spanning a period of at least 3 years and some datasets being extraordinarily long (up to 16 years).

2.2. Estimating soil water content and relative extractable water

Edaphic droughts are characterised by a decrease in the amount of water in the soil. Since locally measured soil water content (SWC) is only available for 56 % of all the SAPFLUXNET datasets and also the depth of SWC measurements is not consistent across datasets, we used SWC [$\text{m}^3 \text{m}^{-3}$] from the ERA5-Land reanalysis product (Muñoz-Sabater et al., 2021) to maximise the coverage and comparability of our analyses. ERA5-Land is characterized by an hourly temporal resolution and a $0.1^\circ \times 0.1^\circ$ (native resolution of 9 km) grid spatial resolution for every day from January 1950 to present. ERA5-Land data has been shown to be a good proxy for locally measured SWC in SAPFLUXNET datasets in previous analyses (Flo et al., 2022). For each of the 202 SAPFLUXNET datasets, we downloaded 40 years (1980–2019) daily time series of SWC at three soil depths: layer 1 (0–7 cm), layer 2 (7–28 cm) and layer 3 (28–100 cm). We then computed a weighted mean (by depth) between layers 1 and 2 to obtain the shallow water content ($\text{SWC}_{0.28}$), which we use in our main analyses. This is motivated by the fact that most roots are expected to be located within the upper 30 cm of the soil (Schenk and Jackson, 2002). However, due to uncertainty regarding the most relevant soil depth to assess water availability to plants, we also computed a weighted mean including all three layers (1, 2 and 3) to calculate water content down to 100 cm ($\text{SWC}_{0.100}$). This variable was used in additional analyses (reported in the Supplementary Material, Tables S9–S14) to assess the consistency of our results.

In order to obtain a comparable estimate of SWC temporal dynamics across sites, we calculated the daily relative extractable water (REW) (Granier et al., 1999), as:

$$\text{REW} = \frac{(\text{SWC} - \text{Q1}(\text{SWC}))}{(\text{Q99}(\text{SWC}) - \text{Q1}(\text{SWC}))} \quad (1)$$

where SWC is the soil water content on a given day, and Q1(SWC) and Q99(SWC) are 1 % and 99 % quantiles of SWC in the 40-year time series, considered to be robust estimates of minimum and maximum SWC, respectively. We used $\text{SWC}_{0.28}$ for the main analyses and $\text{SWC}_{0.100}$ for the Supplementary analyses. REW values should range between 0 and 1, but since we use quantiles, a few SWC values were smaller than the Q1 (SWC) or greater than the Q99(SWC). To avoid this, all REW values lower than 0 were taken as 0 and all REW values higher than 1 were taken as 1. Once we obtained the daily REW time series, we computed a 7-day moving average of REW using the “roll_mean” function from the RcppRoll R package (Ushey, 2013). We did this to smooth out abrupt changes in SWC from one day to the next and to avoid artefacts in drought identification (see next section). This smoothed REW time series was used to define droughts. Finally, we also used REW to calculate a cumulative relative water deficit (RWD) for the drought period in order to have a variable that accounts for both drought intensity and drought length. RWD was calculated as $\text{RWD} = \sum(0.4 - \text{REW})$, where REW is the relative extractable water on a given day, 0.4 is the REW value applied in the following section as a threshold between the pre-drought and the post-drought periods, and $\sum$ is the sum across the whole drought period.

2.3. Drought definition and identification

To identify drought periods with potential impacts on tree functioning, we established a set of criteria to ensure that droughts were comparable to each other and had clear pre-drought and post-drought periods to evaluate water use responses. If a certain period met all the following requirements, it was identified as a drought:

1. At least 10 days of continued REW depletion (decreased REW between consecutive days), indicative of a sustained decrease in soil water availability.
2. A minimum $\text{REW} \leq 0.3$ and an overall REW decrease ≥ 0.2 during the drought, to ensure a substantial soil water depletion.
3. At least 10 days before the drought onset with a mean $\text{REW} \geq 0.4$ (pre-drought period).
4. After reaching a $\text{REW} \geq 0.4$, at least 10 days with a mean $\text{REW} \geq 0.4$ (post-drought period).

The criteria above define the overall extent of a drought. However, small increases in REW (below 0.4), can still occur within the drought period. To avoid potential influences of these conditions on resilience estimation, only days with $\text{REW} \leq 0.3$ (strict drought) were used for the calculation of the resilience metrics (see next section). On account of this, each identified drought event comprises different periods: a 10 day pre-drought period, a 10 day post-drought period, a drought period (*sensu lato*) that lies between the pre- and the post-drought periods, and a strict-drought period that lies within the drought period and has a REW value below 0.3 (Fig. 1).

We recognize that REW thresholds defining soil water limitations in tree water use may vary across biomes and species, but accounting for these variations is difficult in practice. Using REW, however, offers a more comparable drought metric across sites than SWC. The threshold of $\text{REW} < 0.4$, supported by several studies based on sap flow and eddy covariance (Granier et al., 2007, 2000), has been consistently linked to drought limitation across biomes. We used a stricter REW threshold of 0.3 for the drought period, which ensures that trees are experiencing soil water limitations.

2.4. Estimating resilience and its components

Our estimation of tree water use resilience to drought and its components is based on the framework proposed by Lloret et al. to estimate tree growth resilience (Lloret et al., 2011). A recent study has proposed a similar approach, based on the comparison of sap flow before, during and after a drought (Haberstroh and Werner, 2022). As in other drought recovery studies (Skelton et al., 2023, 2017), we propose to use tree canopy conductance (G_c) to make our analyses less dependent on evaporative demand, which could differ substantially between the pre-drought, the drought, and the post-drought periods. We calculated tree-level canopy conductance per unit of sapwood area (or total tree-level canopy conductance, when sapwood area was not available) from sap flow and in situ VPD and air temperature data from SAPFLUXNET, based on Flo et al. (2021). For a visual example of the temporal dynamics of G_c and REW during a typical drought, see Fig. 1. We then aggregated tree canopy conductance for the pre-drought (G_{c_pre}), post-drought (G_{c_post}) and strict-drought (G_{c_dr}) periods. We estimated water use resistance (R_t) as the canopy conductance during the strict-drought relative to the value during the pre-drought period ($R_t = G_{c_dr}/G_{c_pre}$), recovery (R_c) as the canopy conductance during the post-drought period relative to the value during the strict-drought ($R_c = G_{c_post}/G_{c_dr}$), and resilience (R_s) as the canopy conductance during the post-drought period relative to the value during the pre-drought period ($R_s = G_{c_post}/G_{c_pre}$).

The SAPFLUXNET database contains some gaps in the time series of sap flow and meteorological data. To ensure our analyses were robust, we computed the canopy conductance for a given drought event and tree

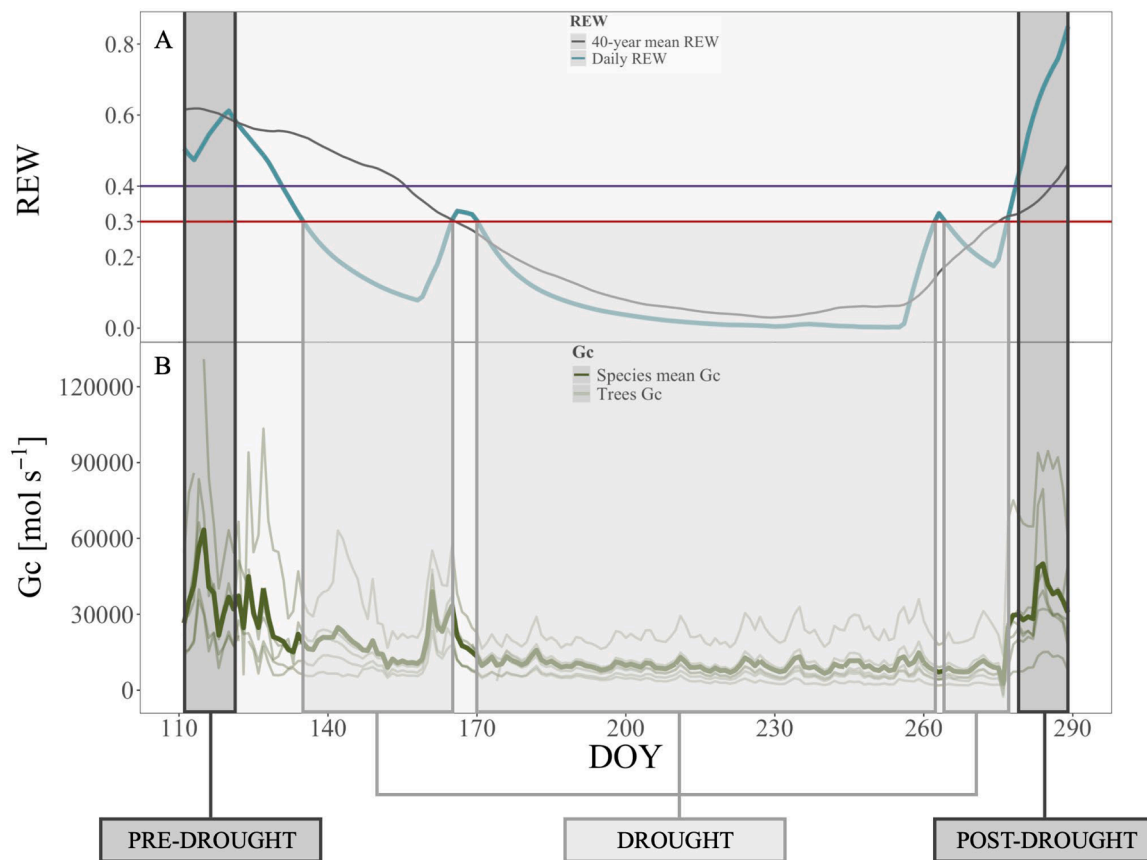


Fig. 1. Time courses of soil drought expressed as the relative extractable water (REW) (A) and whole tree canopy conductance (Gc) (B) to illustrate the criteria for drought identification. Data from Majadas del Tietar (Spain) in 2015 is used as an example. Thick blue line depicts daily REW for 2015, while thin grey line depicts the mean REW for that day of the year (DOY) in the 40-years series (1980–2019). Fine green lines depict canopy conductance for each tree, while the thick green line depicts the mean for all trees of the same species (*Quercus rotundifolia*). Straight red line indicates a REW value of 0.3, below which a drought period occurs. Straight purple line indicates a REW value of 0.4, above which the pre- and post-drought periods are defined. Notice that, when the REW achieves a value of 0.4 the drought comes to an end and the post-drought period begins.

(tree-drought event) only when it had at least 70 % of the days with available data for the pre-, post- and strict-drought periods. After removing tree-drought observations that did not meet this criterion, we ended up with 2001 tree-drought observations including 180 droughts, 89 sites (datasets), 96 species and 909 individual trees. This final dataset, used in all our models, comprises tree-drought observations with a minimum of 80 % coverage of subdaily timestamps, with 93 % of the observations exhibiting complete (100 %) subdaily coverage. This 80 % is a threshold that has been used in recent work at the ecosystem scale (Anderegg et al., 2024). Additional information on the sites and species that we used in our final dataset is provided in the Supplementary Material, Tables S1 and S2, respectively.

2.5. Resilience drivers

To analyse the role of drought characteristics, climate, site factors and plant attributes in the observed variability of resilience metrics, we fitted one linear mixed-effects model for each resilience metric, in which the response variable was the R_t , the R_c or the R_s of each tree-drought event as described above. To achieve normality, all resilience metrics were transformed by using the $\log(\text{variable} + 0.01)$.

The explanatory variables included drought attributes: average REW for the pre- and post-drought periods, the cumulative relative water deficit (RWD) for the drought period, the VPD [kPa] for the pre-drought, drought and post-drought periods, the days since the last previous drought, and the season (Winter, Spring, Summer, Fall; defined in relation to the solstices and equinoxes) when the current drought

occurred. We also included long-term climatic variables: mean annual precipitation (MAP) [m^2] and mean annual temperature (MAT) [$^{\circ}\text{C}$], both obtained from the CHELSA climate database (Karger et al., 2017); and soil properties: the proportion of clay particles in the fine earth fraction [%], the depth to bedrock (up to 200 cm) and the total nitrogen [g kg^{-1}], all directly retrieved from SoilGrids 2.0 (Hengl et al., 2017). Stand-level variables included leaf area index (LAI), obtained from SAPFLUXNET (with 27 site data gaps filled with MODIS), and canopy height [m], again obtained from SAPFLUXNET (with 5 site data gaps filled with the Global 1 km Forest Canopy Height raster (Simard et al., 2011)). We also tested stand density and basal area as alternative variables to characterize stand structure and competition instead of LAI, but none of these variables was selected in the final models. Finally, tree-level variables included the diameter at breast height (DBH) [cm] and the functional group [angiosperms / gymnosperms], all obtained from SAPFLUXNET. To achieve normality of explanatory variables, RWD, VPD, MAP, bedrock depth, nitrogen and DBH were log-transformed; and days since last drought was sqrt-transformed; while other variables were used without transformation. All these predictive variables were chosen based on theory and previous evidence. Additional details on all these variables and the corresponding data sources can be found in Table 1.

In all linear mixed-effects models, species identity nested within site (i.e. 1 | site/species) was included in the random part of all models to account for the hierarchical structure of the data. We initially fitted models including all the explanatory variables described in the previous paragraph (Table 1), as well as the interactions between REW/RWD and

Table 1

Explanatory variables used in the general models of tree water use resilience, resistance and recovery. We included relative extractable water (REW) and vapour pressure deficit (VPD) as drought characteristics (Drought charac.) for the pre- and post-drought periods; relative water deficit (RWD) and VPD for the drought period, as well as days since last drought (Days last drought) and season. Mean annual precipitation (MAP), mean annual temperature (MAT), percentage of soil clay (Clay), bedrock depth (Bedrock) and soil nitrogen concentration were employed to characterise climate and soil, respectively. Leaf area index (LAI), canopy height, diameter at breast height (DBH) and tree functional group (Group) were used to characterise plots and trees, respectively. Note that some variables were not selected in the final models.

Variable type	Name in model	Units	Transform	Median	Range	Description	Source
Drought charac.	Pre-drought	REW	—	0.59	0.40–0.95	Mean Relative Extractable Water during the pre-drought period.	Estimated from ERA5-Land
	Drought	VPD	log0	0.59	0.22–1.77	Mean Vapour Pressure Deficit during the pre-drought period.	SFN
	Drought	RWD	log0	24.82	7.15–298.39	Cumulative Relative Water Deficit, computed as the summation of 1 - REW for the period comprised between the pre- and post-drought periods.	Estimated from ERA5-Land
	Post-drought	VPD	log0	1.31	0.33–5.45	Maximum Vapour Pressure Deficit reached during the strict drought period (REW < 0.3).	Estimated from ERA5-Land
	Post-drought	REW	—	0.61	0.40–0.87	Mean Relative Extractable Water during the post-drought period.	SFN
Climate	Days last drought	VPD	log0	0.61	0.18–1.50	Mean Vapour Pressure Deficit during the post-drought period.	See methods
	Days last drought	N°	sqrt0	274	5–1444	Days passed since the last identified drought at the same site.	SFN
	Season	—	—	—	—	Winter / Spring / Summer / Fall.	SFN
	MAP	kg m ⁻²	log0	1202.54	248–3625.91	Mean Annual Precipitation at the site.	CHELSA
	MAT	°C	—	11.69	−0.98–28.90	Mean Annual Temperature at the site.	CHELSA
Soil	Clay	%	—	29.30	3.70–44.60	Percentage of clay in the soil.	SFN - SoilGrids
	Bedrock	cm	log0	1943	329–12,030	Soil bedrock depth.	SFN - SoilGrids
	Nitrogen	g kg ⁻¹	log0	1.52	0.23–14.26	Soil nitrogen concentration.	SFN - SoilGrids
Plot characteristics	LAI*	m ²	—	4.90	0.22–7.03	90th quantile of the Leaf Area Index as a measure of the maximum value.	SFN - MODIS
	Canopy height	m	—	20	0–45	Canopy height.	SFN - FCHr
Tree characteristics	DBH	cm	log0	28.76	2.15–220	Diameter at Breast Height for each tree.	SFN
	Group	—	—	—	—	Angiosperms / gymnosperms.	SFN

* Stand density and basal area were also considered as alternative descriptors of stand structure instead of LAI (as explained in Section 2.5).

VPD for each corresponding drought period (i.e., REW (pre) * VPD (pre), RWD (drought) * VPD (drought) and REW (post) * VPD (post)). Then, models were simplified stepwise and we selected the one with the lowest Akaike's information criterion (AIC) (see Supplementary Material, Tables S3–S5), as our aim was to identify significant effects, not maximise prediction efficiency *per se*. These simplified general models are the ones presented in our Results (Table 2 and Figs. 2–5).

2.6. Species traits influence on resilience

We conducted an additional set of analyses to assess whether species traits predicted resilience metrics. To this end, we first fitted a model for each resilience metric that was identical to the ones described above, except for the functional group variable, which was not included. This was done because many of the traits studied are known to differ between angiosperms and gymnosperms, and including this variability *a priori* would have partialled out this effect. We extracted the random coefficients from these alternative models and used them as aggregate measures of each resilience metric for each species-site combination. These coefficients were used as response variables in a new set of models in which we assessed the effect of species traits on resilience metrics. Note that, the effects of the potential drivers included in the general model (see previous section) are accounted for in our trait analyses, as the species-level coefficients we extracted correspond to the species effect once the effect of all other explanatory variables in the general model has been controlled for.

We selected traits that are known to be involved in plant drought responses or general resource-use strategies, including wood density (WD), specific leaf area (SLA), the water potential causing 50 % loss of xylem hydraulic conductivity due to embolism (P50), the turgor loss point (TLP) as a proxy for stomatal control of water use under drought (Martin-StPaul et al., 2017), the Huber value (Hv), the hydraulic safety margin (HSM) and the stem-specific hydraulic conductivity (Ks). All these traits were obtained from previously published datasets (Flo et al., 2021; Mencuccini et al., 2019; Sanchez-Martinez et al., 2020). The HSM was defined as the difference between the minimum leaf water potential and P50, but we also considered alternative definitions using P88 for angiosperms and P50 for gymnosperms, as suggested by some authors (Urli et al., 2013). We fitted one separate model for each of these traits, due to constraints in sample size. Not all traits were available for all species, and hence sample size for the trait models varied between 20 and 48 species: P50 (40 species), Ks (34), TLP (34), Hv (32), HSM (20), SLA (45), WD (48). Considering that we had several estimates for some species (due to multiple sampling locations, 93 - 135 individual observations per model), we included species as a random effect (1 | species) in these trait models. In addition, we assessed the potential effect of functional groups [angiosperms / gymnosperms] and leaf habit [deciduous / evergreen] by including an interaction between these latter variables and each trait. Again, we started with the more complex model contemplating the interactions (trait * functional group or trait * leaf habit) and followed a stepwise process until we found the model with the best fit in terms of AIC. We depicted the functional group in the main analyses and the leaf habit in the Supplementary Material (Table S15). SLA was log-transformed to achieve normality, and all other variables were included without transformation.

All the statistical analyses were performed using the R software version 4. Specifically, linear mixed-effects models (LMM) were fitted with the lmer() function, and model selection was conducted using the step() function; both available in the lmerTest R package (Kuznetsova et al., 2017). To check linearity, homogeneity of variances and influential observations and to check for variance inflation factors in complex models we used the check_model() function from the performance R package (Lüdtke et al., 2021).

Table 2

Summaries of the significant effects (p, for probability, significant values in bold font) of all the tested explanatory variables on resilience, resistance and recovery (log-transformed), according to the mixed-effects models. Functional group explanatory variable is depicted as 'group' for better clarity.

Variable			Resilience			Resistance			Recovery		
Variable type	Predictors		Est.	(CI)	p	Est.	(CI)	p	Est.	(CI)	p
Drought charac.	(Intercept)		0.42	(0.18–0.67)	0.001	0.11	(–0.09–0.31)	0.272	0.12	(–0.02–0.25)	0.098
	Pre-drought	REW	0.04	(0.01–0.08)	0.021	0.07	(0.04–0.10)	<0.001	–0.05	(–0.08–0.02)	0.002
		REW * group	0.14	(0.09–0.19)	<0.001	–0.04	(–0.09–0.00)	0.039	0.21	(0.17–0.26)	<0.001
		VPD	0.24	(0.19–0.29)	<0.001	0.14	(0.10–0.18)	<0.001	0.11	(0.07–0.14)	<0.001
	Drought	VPD * group	0.11	(0.04–0.18)	0.003	0.15	(0.10–0.21)	<0.001	—	—	—
		REW * VPD	–0.04	(–0.07–0.01)	0.004	0.02	(0.00–0.04)	0.044	–0.08	(–0.11–0.06)	<0.001
		RWD	–0.11	(–0.16–0.07)	<0.001	0.13	(0.10–0.17)	<0.001	–0.24	(–0.28–0.20)	<0.001
	Post-drought	RWD * group	—	—	—	—	—	—	—	—	—
		VPD	–0.01	(–0.06–0.04)	0.726	–0.19	(–0.23–0.14)	<0.001	0.16	(0.10–0.21)	<0.001
		VPD * group	—	—	—	–0.17	(–0.23–0.11)	<0.001	0.23	(0.16–0.31)	<0.001
	Season	RWD * VPD	–0.05	(–0.08–0.02)	0.002	–0.06	(–0.09–0.04)	<0.001	0.05	(0.02–0.08)	0.002
		REW	0.12	(0.08–0.16)	<0.001	NA	NA	NA	0.07	(0.03–0.11)	<0.001
		REW * group	–0.13	(–0.19–0.07)	<0.001	NA	NA	NA	–0.13	(–0.18–0.08)	<0.001
	days last drought	VPD	–0.11	(–0.15–0.06)	<0.001	NA	NA	NA	–0.06	(–0.10–0.01)	0.009
		VPD * group	–0.14	(–0.21–0.07)	<0.001	NA	NA	NA	–0.26	(–0.32–0.20)	<0.001
		REW * VPD	0.07	(0.04–0.10)	<0.001	NA	NA	NA	0.04	(0.02–0.07)	0.002
	days last drought		0.19	(0.15–0.22)	<0.001	0.07	(0.04–0.10)	<0.001	0.09	(0.07–0.12)	<0.001
	days last drought * group		–0.10	(–0.16–0.04)	0.001	–0.04	(–0.09–0.00)	0.037	—	—	—
	Season	Spring	–0.32	(–0.55–0.10)	0.004	–0.09	(–0.27–0.10)	0.345	—	—	—
		Summer	–0.52	(–0.73–0.30)	<0.001	–0.34	(–0.51–0.16)	<0.001	—	—	—
		Fall	–0.61	(–0.82–0.41)	<0.001	–0.43	(–0.59–0.26)	<0.001	—	—	—
Climate	MAP		—	—	—	—	—	—	—	—	—
	MAT		—	—	—	–0.11	(–0.18–0.04)	0.003	0.12	(0.03–0.22)	0.012
Soil	Bedrock		—	—	—	–0.09	(–0.17–0.00)	0.041	—	—	—
	Nitrogen		—	—	—	—	—	—	0.15	(0.05–0.24)	0.004
Tree characteristics	DBH		0.05	(0.02–0.09)	0.002	—	—	—	0.04	(0.01–0.07)	0.018
	Group	Gymnosperms	–0.05	(–0.25–0.15)	0.633	–0.19	(–0.31–0.07)	0.003	–0.04	(–0.25–0.17)	0.730

3. Results

3.1. Effect of drought characteristics on resilience

Rs was promoted by high soil water availability (REW) during the pre- and the post-drought periods, and also by high soil water availability (low RWD) during the drought period (Table 2, Fig. 2). Higher VPD during the pre-drought period promoted Rs, but the opposite effect of VPD was observed during the post-drought period (Table 2, Fig. 2). The effects of both REW and VPD interacted with the species functional group, with steeper responses for gymnosperms except for post-drought REW. In addition, the interaction between REW and VPD was significant for all the pre-, drought and the post-drought periods. Before the drought, the REW effect turned from positive to negative as VPD increased (Fig. 2G), while for the drought period the negative effect of RWD became stronger as VPD increased (Fig. 2H), and for the post-drought period the positive effect of REW became steeper as VPD increased (Fig. 2I). The time since the last (previous) drought also had an effect on resilience, with longer times implying higher Rs, particularly for angiosperms (Table 2, Fig. 3A). Finally, the season had also a significant effect on resilience, with lower values of Rs associated with droughts occurring in the spring, summer or fall, relative to those occurring in winter (Table 2, Fig. 3B). Full model results, including random effects (site and species within site) are provided in Supplementary Material, Tables S6–S8.

As expected due to their non-independent nature, the effects of a given explanatory variable on Rt and Rc tended to be inverse (Table 2). In most cases, the effects of the study variables on Rs were driven by their effect on Rt rather than Rc (Table 2, Figs. 4 and 5). Exceptions to this rule involve the interactions of pre-drought REW with functional group and with pre-drought VPD, the effect of RWD, post-drought effects, and the effect of DBH. Note that post-drought effects can not affect Rt by definition. For plots on Rc significant effects see Supplementary Material, Figures S1 and S2.

3.2. Effects of site, climate, soil, stand and tree-level characteristics

Rs was not explained by site-level factors such as mean annual temperature (MAT), mean annual precipitation (MAP), soil nitrogen or depth to bedrock (Table 2). However, a warmer climate was associated with a lower Rt (Table 2, Fig. 5C) and a higher Rc (Table 2). Among the soil characteristics, a higher nitrogen availability was associated with higher Rc (Table 2), while, surprisingly, Rt was lower in deeper soils although this effect of bedrock depth is not observed for Rs (Table 2, Fig. 5D). Stand characteristics such as LAI and canopy height did not influence any resilience metric, as these predictors were never selected in the models after applying the step function (Table 2; see also Supplementary Material, Tables S3–S5). Among the tree characteristics, trees with a larger diameter at breast height (DBH) showed increased tree water use Rs (Table 2, Fig. 3C) and Rc (Table 2). Lastly, gymnosperms showed a lower Rt in comparison to angiosperms (Table 2).

3.3. Effects of species and traits on resilience

Rs increased with higher (i.e., less negative) P50 and lower (more negative) TLP, in both cases regardless of the species functional group (Table 3, Fig. 6A,B). WD was negatively related to Rs in gymnosperms but we could not detect an effect of WD on Rs for angiosperms (significant WD*functional group interaction; Table 3, Fig. 6C). The other traits examined (Ks, SLA, Hv and HSM) did not show any effect on Rs (Table 3). Trait effects on Rs remained similar when we used leaf habit as a covariate in the models instead of functional group, with the exception that the positive effect of P50 was only significant for evergreen species, and that no effect was observed for WD (Table S15).

Rt, similar to Rs, also increased in species with a higher P50 and a lower TLP. In the last case, the functional group variable was also significant. The HSM tended to be positively related to Rt in gymnosperms, but the effect was only marginally significant ($p = 0.076$) (Fig. 7E). None of the results regarding the HSM changed if this variable was estimated using the P88 instead of P50 for angiosperms (data not shown). Rt was not affected by WD, SLA or Hv, and only varied as a function of species

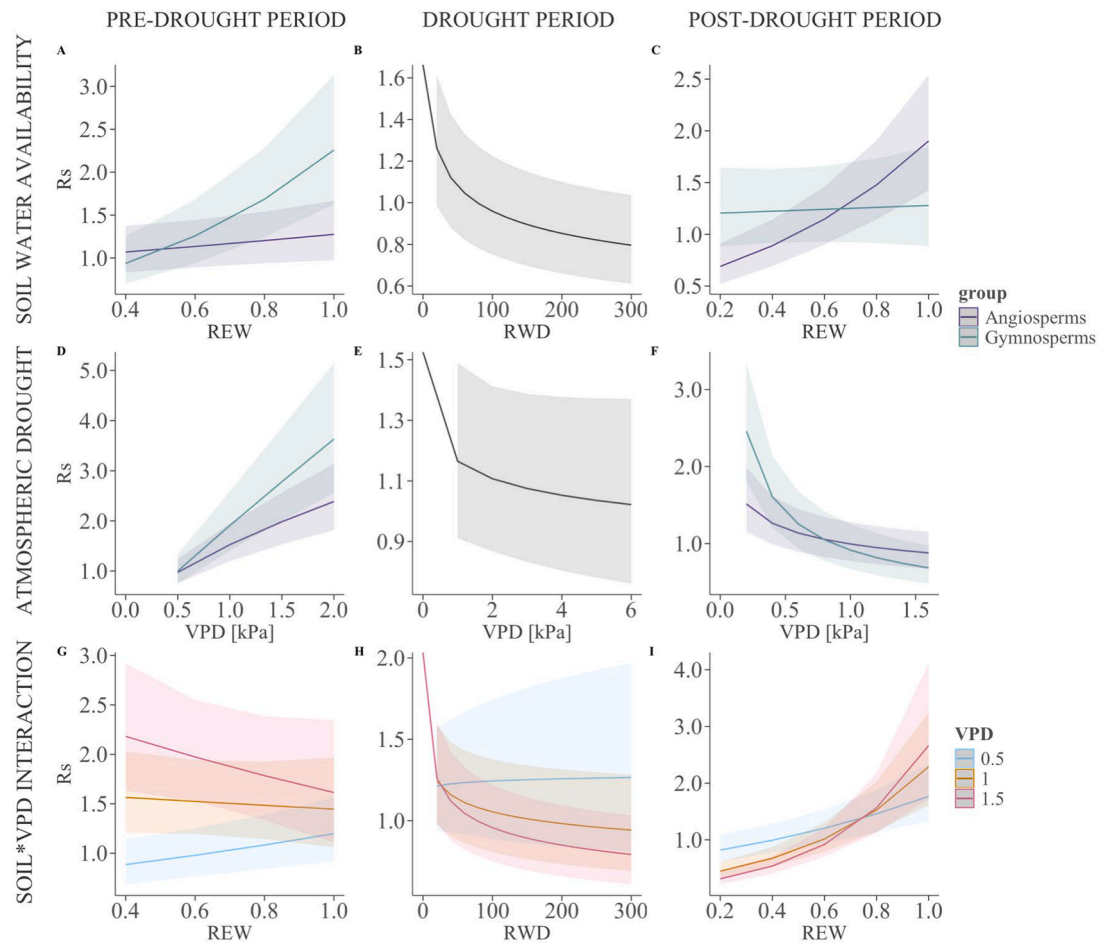


Fig. 2. Modelled effects and 95 % confidence intervals of drought characteristics on water use resilience (R_s). Drought variables include the soil moisture conditions (relative extractable water, REW; or relative water deficit, RWD; both dimensionless) and the vapour pressure deficit (VPD [kPa]) during the pre-drought, drought and post-drought periods. Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Full model results, including exact p -values for each effect, are reported in Table 2.

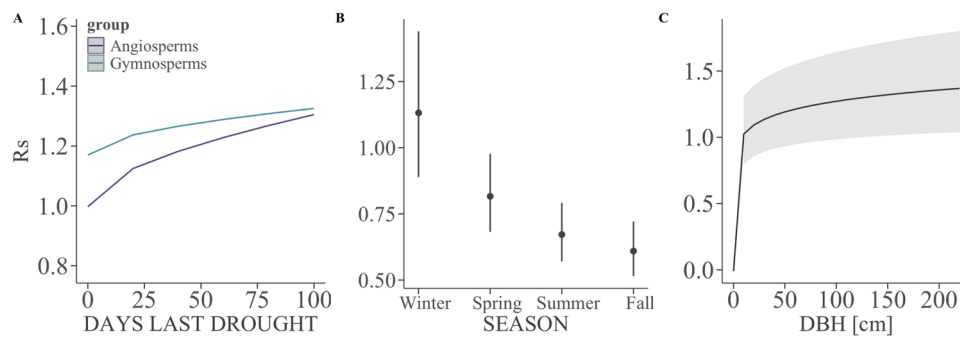


Fig. 3. Modelled effects and 95 % confidence intervals of climate, soil, plot and tree characteristics on water use resilience (R_s). Variables include days since the last previous drought (A), season in which the drought took place (B) and diameter at breast height [cm] (DBH) (C). Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Confidence interval is not shown for panel A for better visibility. Full model results are reported in Table 2.

functional group in the corresponding models (Table 3, Fig. 7). Trait effects on R_t were similar when we used leaf habit as a covariate in the models instead of functional group, except for a significant positive effect of WD for deciduous species (Table S15). R_c was not affected by any of the traits studied, neither alone nor in interaction with the species functional group.

4. Discussion

Our main objective was to identify the main biotic and abiotic drivers that determine the observed patterns of tree water-use resilience to drought. In the following sections we discuss how drought characteristics, site-level environmental factors and species traits determine the observed resilience responses, emphasizing the consistency (or lack thereof) of the observed patterns with our initial hypotheses.

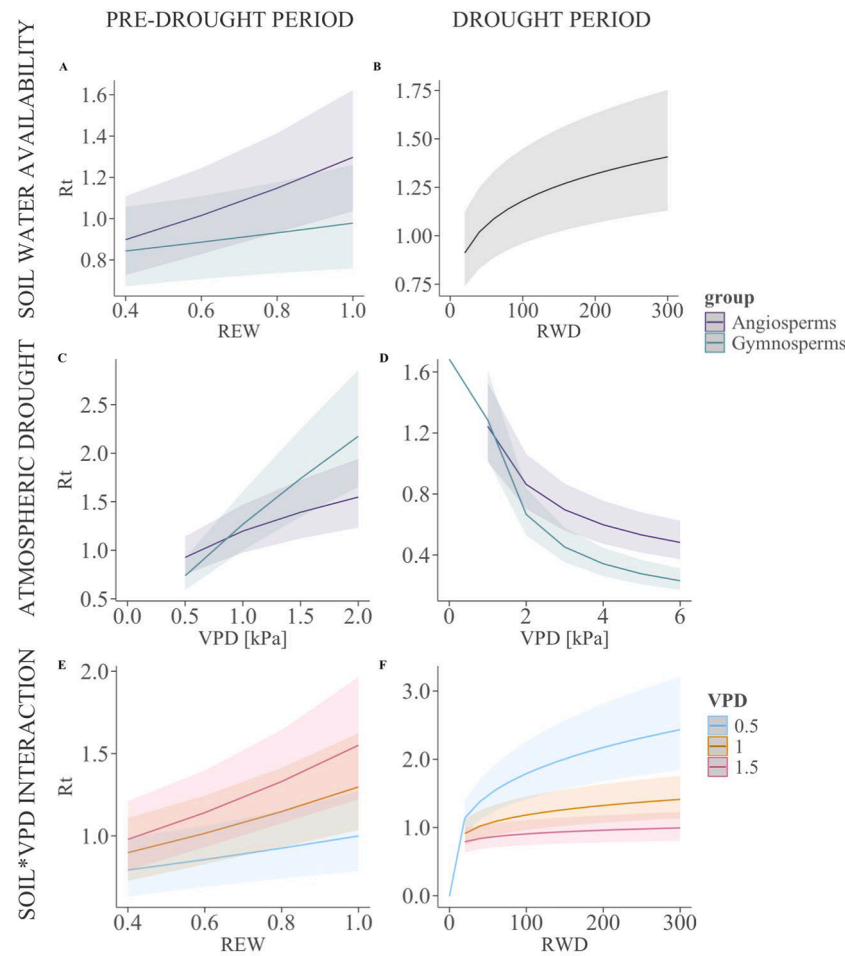


Fig. 4. Modelled effects and 95 % confidence intervals of drought characteristics on water use resistance (R_t). Drought variables include the soil moisture conditions (relative extractable water, REW; or relative water deficit, RWD; both dimensionless) and the vapour pressure deficit (VPD [kPa]) during the pre-drought and drought periods. Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Full model results are reported in Table 2.

4.1. Soil water availability and atmospheric demand determine tree water use resilience to drought

We hypothesised that higher soil water availability and milder atmospheric drought conditions before, during, and/or after the drought will promote tree water use resilience. Our findings confirmed that R_s was indeed promoted by moist soils (higher REW) before, during, and after the drought. R_s also increased with decreasing atmospheric demand after the drought (Table 2, Fig. 3). These results are consistent with several global and regional studies assessing resilience from remotely-sensed vegetation indices (i.e., Jiao et al., 2021; Yao et al., 2023). Across terrestrial ecosystems globally, soil moisture during the recovery period was the dominant driver of resilience, followed by evaporative demand (Yao et al., 2023). It is important to note that our predictor of soil water deficit during drought (RWD), negatively related to resilience, includes both drought duration and intensity, which have also been reported to reduce resilience across Australian ecosystems (Jiao et al., 2021). Interestingly, the interaction between VPD and soil moisture in determining R_s was significant for all three pre-, drought and post-drought periods (VPD*REW and VPD*RWD interactions) (Table 2, Fig. 3). The negative direction of this interaction in the drought period (where we use RWD) and the positive direction in the post-drought period (where we use REW) was as expected, as R_s decreased when both the soil and the atmosphere were dry. However, in the pre-drought period, we found that high atmospheric demand for water promoted R_s , contrary to our expectation. The most likely

explanation for this finding is that high pre-drought VPD reduces tree canopy conductance (Oren et al., 1999), which lowers the reference state for our resilience calculations and thus facilitates reaching similar or even higher conductance values in the post-drought period. The fact that VPD is directly involved in the estimation of canopy conductance implies that the VPD effects observed in this study should be interpreted with caution.

Recurrent droughts, i.e. shorter time intervals since the last previous drought, negatively affected R_s (Table 2, Fig. 3A), as we hypothesised. These results are consistent with those obtained in the same study on Australian ecosystems mentioned above, albeit that study focused on droughts defined at a longer time scale (i.e. monthly as opposed to daily here); in that case, periods of three months or less between droughts greatly hampered full recovery (Jiao et al., 2021). What are the main biological mechanisms that could explain this decreased resilience following recurrent droughts? Firstly, a cumulative hydraulic damage in the xylem that leads to an impaired hydraulic system that ultimately cannot be recovered (Brodribb and Cochard, 2009; Delzon and Cochard, 2014). Secondly, potential losses in hydraulic conductance along the entire soil-to-leaf pathway (Ruehr et al., 2019), in addition to those caused by xylem embolism. Outside-xylem hydraulic conductance losses (leaves, roots) that occur early in droughts may not recover fully and may explain the decreased resilience (Skelton et al., 2023, 2017). Furthermore, trees that experience frequent droughts may have depleted carbohydrate reserves and reduced root growth, along with the deteriorated hydraulic systems (Anderegg et al., 2020, 2015; Brodribb et al.,

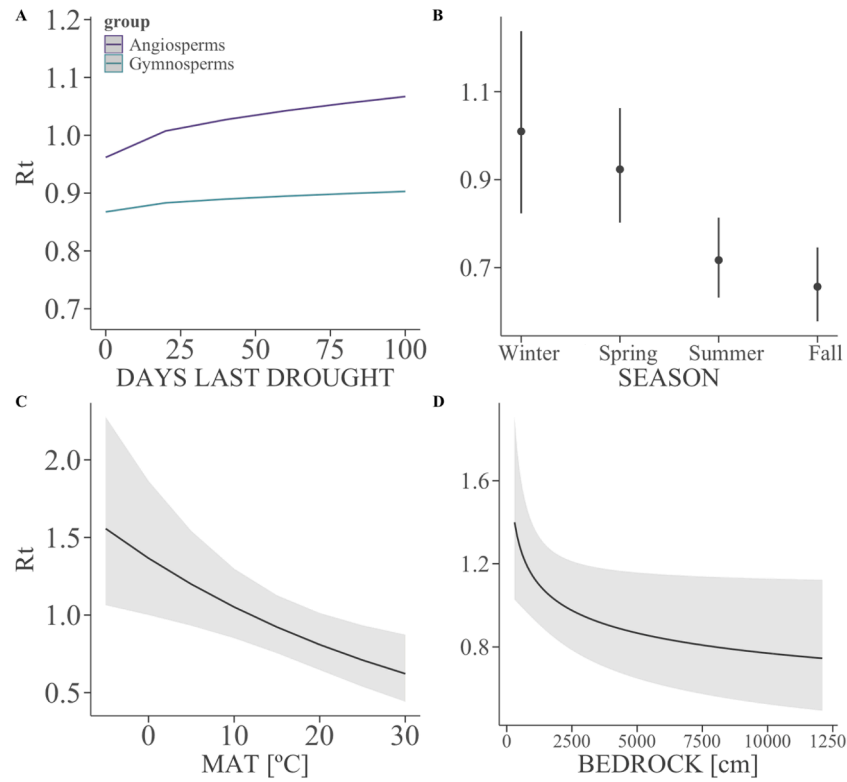


Fig. 5. Modelled effects and 95 % confidence intervals of climate, soil, plot and tree characteristics on water use resistance (R_t). Variables include days since the last previous drought (A), season in which the drought took place (B), mean annual temperature [$^{\circ}\text{C}$] (C) and depth to soil bedrock [cm] (D). Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Confidence interval is not shown for panel A for better visibility. Full model results are reported in Table 2.

Table 3

Summaries of the significant trait effects on species-level resilience, resistance and recovery, according to the mixed-effects models. Variables include the effect of vulnerability to embolism (P50) [MPa], the stem-specific hydraulic conductivity (Ks) [$\text{kg m}^{-1} \text{MPa}^{-1} \text{s}^{-1}$], the turgor loss point (TLP) [MPa], the Huber value (Hv) [$\text{cm}^2 \text{m}^{-2}$], the hydraulic safety margin (HSM) [MPa], the specific leaf area (SLA) [$\text{m}^2 \text{kg}^{-1}$], and wood density (WD) [g cm^{-3}]. The interaction of every trait with the functional group [angiosperms / gymnosperms] (Group) was also assessed.

Variable		Resilience			Resistance			Recovery		
Predictors		Est.	(CI)	p	Est.	(CI)	p	Est.	(CI)	p
P50 (40 species, $n = 126$)	P50	0.06	(0.00–0.11)	0.047	0.04	(0.01–0.07)	0.003	—	—	—
	Group	0.02	(–0.08–0.13)	0.652	—	—	—	—	—	—
	P50 * group	—	—	—	—	—	—	—	—	—
Ks (34 species, $n = 111$)	Ks	—	—	—	—	—	—	—	—	—
	Group	—	—	—	—	—	—	—	—	—
	Ks * group	—	—	—	—	—	—	—	—	—
TLP (34 species, $n = 107$)	TLP	–0.06	(–0.11–0.01)	0.022	–0.04	(–0.08–0.01)	0.008	—	—	—
	Group	—	—	—	–0.08	(–0.16–0.01)	0.034	—	—	—
	TLP * group	—	—	—	—	—	—	—	—	—
Hv (32 species, $n = 93$)	Hv	—	—	—	–0.00	(–0.06–0.06)	0.984	—	—	—
	Group	—	—	—	–0.13	(–0.27–0.00)	0.047	—	—	—
	Hv * group	—	—	—	—	—	—	—	—	—
HSM (20 species, $n = 94$)	HSM	—	—	—	–0.03	(–0.08–0.01)	0.140 ⁺	—	—	—
	Group	—	—	—	–0.08	(–0.16–0.01)	0.079	—	—	—
	HSM * group	—	—	—	0.14	(0.01–0.26)	0.029	—	—	—
SLA (45 species, $n = 127$)	SLA	—	—	—	–0.01	(–0.06–0.03)	0.551	—	—	—
	Group	—	—	—	–0.14	(–0.25–0.04)	0.007	—	—	—
	SLA * group	—	—	—	—	—	—	—	—	—
WD (48 species, $n = 135$)	WD	0.01	(–0.07–0.10)	0.772	0.02	(–0.02–0.05)	0.347	—	—	—
	Group	–0.25	(–0.50–0.00)	0.047	–0.08	(–0.16–0.00)	0.045	—	—	—
	WD * group	–0.34	(–0.63–0.05)	0.023	—	—	—	—	—	—

⁺ Note that the direct effect of HSM for Gymnosperms was not significant either ($p = 0.076$).

2020), all of which can weaken their ability to recover and maintain water use.

Finally, R_s was lower when the droughts took place during the warm summer months and the fall season (Table 2, Fig. 3B). Summer and fall

are probably the seasons when soil water reserves are most depleted, especially considering that the SAPFLUXNET database is biased towards Mediterranean and temperate biomes (Poyatos et al., 2021). The fact that we detect these seasonal effects on R_s may suggest that deeper

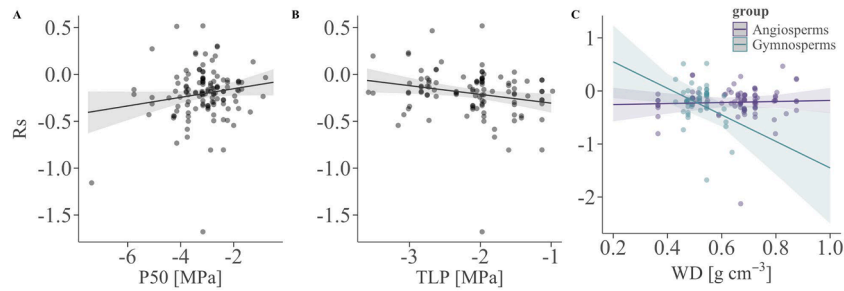


Fig. 6. Modelled effects of plant traits on species-level resilience (R_s), including the effect of vulnerability to embolism (P50) [MPa] (A), the turgor loss point (TLP) [MPa] (B), and the wood density (WD) [g cm^{-3}] (C). Data points correspond to combinations of species and sites. Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Full model results are reported in Table 3.

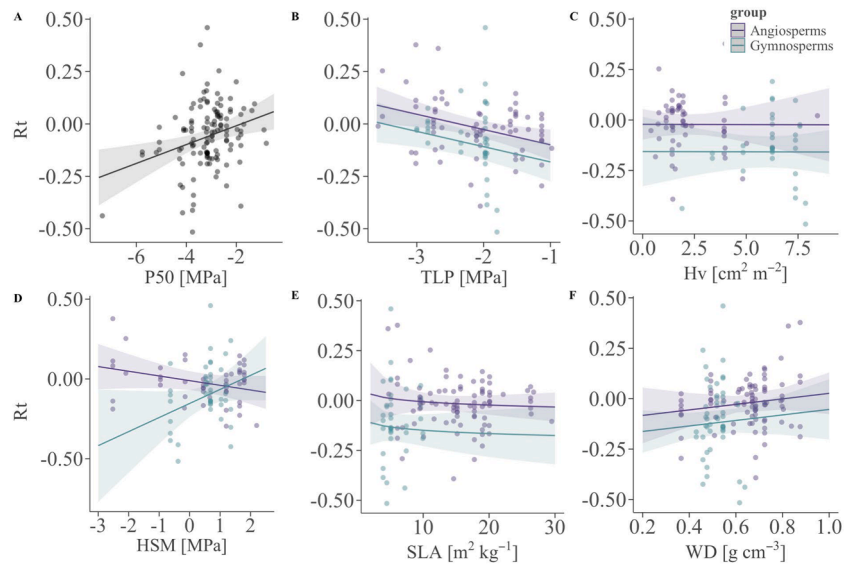


Fig. 7. Modelled effects of plant traits on species-level resistance (R_t), including the effect of vulnerability to embolism (P50) [MPa] (A), the turgor loss point (TLP) [MPa] (B), the Huber value (Hv) [$\text{cm}^2 \text{m}^{-2}$] (C), the hydraulic safety margin (HSM) [MPa] (D), the specific leaf area (SLA) [$\text{m}^2 \text{kg}^{-1}$] (E), and the wood density (WD) [g cm^{-3}] (F). Data points correspond to combinations of species and sites. Only significant effects ($p < 0.05$) are depicted, and the effect of the functional group is shown if significant (see methods). Full model results are reported in Table 3.

soil water reserves, beyond the upper 28 cm used to determine soil water availability in this study, may additionally determine water use resilience. However, we found similar results when accounting for deeper soil water reserves (SWC down to 1 m depth) (see the additional analyses reported in the Supplementary Material, Tables S9-S14). We cannot rule out other phenological influences on our analysis, as we analysed all detected droughts, without filtering out those outside the growing season. In our dataset, leaf areas will probably be lower in winter and (early) spring, reducing whole-tree transpiration demand and facilitating a higher R_s during these seasons.

4.2. Tree water use resilience is not explained by overall climate or soil factors

We expected that sites that were more humid would be characterised by higher tree resilience to drought than more arid sites, but we did not find any effect of overall climate (MAP, MAT) on R_s . Our results differ from previous research, like Hu et al. (2022), who estimated resistance and resilience as the capacity of gross primary productivity (GPP), obtained from eddy covariance data, to withstand drought impacts, and found positive effects of MAP on GPP-based resistance and resilience metrics. Despite the fact that we did not find an effect of MAT on R_s , we should take into account that this variable affected resistance and recovery in opposite directions, possibly resulting in the observed lack of

effect on R_s .

Also contrary to our initial hypothesis, more fertile soils (richer in nitrogen) were not associated with increased R_s to drought. However, they were actually related to enhanced tree water use R_c (Table 2), consistent with the idea that less productive sites generally require longer recovery times than more fertile ones (Yu et al., 2017). During a drought period, shortage of soil nutrients can lead to a weakened plant nutritional status and hence damage its general functioning and resistance to stress (Kreuzwieser and Gessler, 2010). However, the role of nutrients in explaining drought resilience is complex, since high nutrient availability may predispose trees to suffer higher drought impacts, by driving increased biomass allocation to aboveground tissues (e.g., leaf area), promoting hydraulic failure and increasing carbon costs (Gessler et al., 2017). On the other hand, nutrients also boost drought recovery by enhancing water transport capacity and water use efficiency (Gessler et al., 2017).

In contrast to previous studies showing decreased growth-based resilience with height (Camarero et al., 2024; McGregor et al., 2021), we found that large tree diameter (DBH) positively affected R_s (Table 2, Fig. 3C). Size effects can be driven by height-related constraints on water transport and higher evaporative demand of exposed canopies, leading to lower leaf water potentials and higher water stress (Fernández-de-Uña et al., 2023). On the other hand, taller and thicker trees have more sapwood volume where they can store more water,

which helps them draw water out of its tissues when the soil is dry. They may also have a proportionally larger root system, which also allows them to take up more water from deeper soil and phreatic layers (Fernández-de-Uña et al., 2023; Giardina et al., 2018). Our results could thus be explained by the fact that larger trees may have a deeper or more extensive root system and a greater water storage capacity.

4.3. Tree water use resilience is mediated by species-level hydraulic traits

We expected traits associated with less strict stomatal regulation (more negative TLP) or higher hydraulic safety (more negative P50, higher HSM) would confer higher resistance and resilience. Surprisingly, however, we found that species that were more vulnerable to stem xylem embolism (i.e., less negative stem P50) exhibited higher Rs and Rt (Table 3, Figs. 6 and 7). These results may be explained by the fact that hydraulic injury is not directly determined by vulnerability to embolism (as measured by the P50 metric) but by this vulnerability in relation to the water stress experienced by the plant (e.g., the HSM metric, Choat et al., 2012). A recent study, for instance, has shown that GPP resilience decreases in forests dominated by species with low hydraulic safety margins (Hu et al., 2022). The fact that in our case the effect of HSM was not significant on either Rs or Rt suggests that heterogeneous species exposure (i.e., differences in the minimum leaf water potential experienced by each species) compensated for differences in hydraulic vulnerability, partially explaining the counterintuitive effect of P50 that we observed. Our findings align with the positive relationship observed between stem P50 and drought recovery of predawn leaf water potentials by Yin and Bauerle (2017), although their analysis focused on different metrics (recovery, not resilience) and response variables (leaf water potentials) and it is well known that recovery dynamics of gas exchange and water potentials may differ (Ruehr et al., 2019).

As hypothesised, we detected that species with a less negative TLP were also less resilient and resistant to drought (Table 3, Figs. 6 and 7). Tree species with higher (less negative) TLPs tend to close their stomata earlier (Martin-StPaul et al., 2017), which can limit water use, carbon uptake and growth during drought (Bartlett et al., 2012), and hamper metabolic recovery after drought (Ruehr et al., 2019). All these effects would contribute towards lower water-use resilience to drought.

4.4. Relationship of water-use resilience and tree resilience at large spatiotemporal scales

In the introduction, we propose that a more detailed understanding of tree water use resilience in the short-term (weeks to months) is likely to provide key insights into the factors that determine the functional response of forests (e.g., in terms of growth) to more severe and frequent droughts. Here, we highlight the relationships between drought resilience of tree water use and tree resilience at larger temporal and spatial scales (i.e., tree rings and remote sensing data) as reported in previous studies. With regard to drought characteristics, we have shown that trees need adequate soil water conditions to cope with severe stresses and to be able to recover pre-drought canopy conductance, and that drought severity (intensity and duration) decreases resilience. These relationships are less clear in large-scale assessments of resilience based on growth and vegetation indices that examined the effect of drought intensity on resilience (Gazol et al., 2018). Our finding of negative effects of drought recurrence on water use resilience supports recent studies which show that the increasing recurrence of dry months and years decreases growth resistance and resilience to drought (Bose et al., 2020; Gao et al., 2018). We also report an influence of drought timing on resilience, which differs from that observed in growth resilience studies. While we find a lower water use resilience in summer and fall, droughts occurring earlier in the growing season seem to decrease growth resilience because of a stronger effect on current year growth (Gao et al., 2018; Kannenberg et al., 2019).

As expected, we found some influences of traits and functional group

[angiosperms / gymnosperms] on water use resilience or some of its components. Our result that the HSM was not related to resilience contrasts with those found in a recent study on semi-arid conifers, which showed higher HSM associated with higher growth resilience to drought (Duan et al., 2022). This same study reports higher resilience with increasing WD, and also increased growth resistance with Ks, results which are partially consistent with ours. In contrast, Song et al. (2022) did not find hydraulic traits to be related to growth resilience to drought across conifer species and they attributed this lack of effect to the wide hydraulic safety margins. We also detected that species with a less negative TLP were also less resilient and resistant to drought. These results align with those from a study on growth-based resilience of temperate tree species, which has shown that species with earlier stomatal closure or able to maintain less negative leaf water potentials tend to be less resilient and resistant to drought (McGregor et al., 2021). Finally, the generally lower resilience observed for gymnosperms here is consistent with more pervasive drought effects on carbon dynamics in ecosystems dominated by this species group (Anderegg et al., 2015).

Overall, both a short-term focus on water use and a longer-term focus on tree ring resilience highlight the impacts of antecedent and posterior environmental conditions as well as the role of plant traits in affecting tree-scale resilience at different temporal scales. Because old damaged xylem conduits are generally replaced by new functional ones only via xylem growth, the two time scales are coupled. At seasonal time scales, complete hydraulic recovery following droughts in summer and fall must wait for new growth during the next growing season. At inter-annual time scales, reduced water availability during winter and the subsequent spring will reduce tree growth and therefore indirectly impact on the resilience in water use of the subsequent year. We call for systematic studies comparing resilience indices simultaneously at different temporal and spatial scales to help understand the complex link existing between these different processes.

4.5. Caveats

We recognize that the three metrics we used to characterize resilience, Rs, Rt and Rc, are interdependent (Li et al., 2020; Schwarz et al., 2020; Zheng et al., 2021), which is why we focus on Rs. However, we still present each index individually to capture different perspectives of tree responses, as the patterns for Rt and Rc may help understand the underlying mechanisms. Another caveat arises from the calculation of our resilience metrics, which are based on the delimitation of pre- and post-drought periods of a specific duration and the quantification of average tree water conductance over those periods. An alternative, more flexible approach to quantifying resilience could involve the use of model-based baselines to estimate tree physiological performance (i.e. sap flow) in the absence of drought effects and then derive resilience metrics based on the entire sap flow time series (counterfactual approach), as done for ecosystem fluxes (Mathes et al., 2021).

Quantifying soil droughts based on daily data across sites globally is challenging, and we had to resort to global ERA5-Land soil moisture to maximise the number of sites and species in our analysis. Droughts were identified by considering the water volume within the upper soil layer (0 - 28 cm), where the majority of tree roots are located (Bachofen et al., 2024; Perry, 1982; Schenk and Jackson, 2002; Tumber-Dávila et al., 2022). While drought detection is not affected by the chosen depth, this may be a limitation due to some trees accessing deeper phreatic layers and this could indeed affect the soil water predictors, so we validated our study by conducting parallel analyses with a soil layer depth of up to 100 cm. These analyses revealed very similar patterns to those obtained considering only the upper soil layer, and hence we conclude that different definitions of soil depth did not have a strong effect on our results (see Supplementary Material, Tables S9-S14). However, it is also possible that the resolution of the ERA5-land model-data fusion product is probably too coarse to disentangle the influence on water use resilience of soil water at different depths. As any synthesis study of global

scope, our results are limited by multiple data availability and representativity issues. Although our study covers a fairly high number of sites (89) and tree species (96), there is a clear bias towards certain biomes such as Mediterranean and temperate regions, thus limiting the representativity for other biomes, particularly tropical regions (only 6 sites). Seasons were defined based on the solstices and equinoxes, and while we acknowledge that this definition may not be meaningful in some cases (e.g., tropical sites), we need to account for the fact that drought resilience may be season-dependent in Mediterranean and temperate regions (Forner et al., 2018), which dominate our dataset. The overlap between SAPFLUXNET and the trait databases was partial, so that hydraulic trait analyses included only 43 species on average. This highlights the need to expand such databases with more sites, species and traits to enhance global representativity in future cross-biome syntheses of tree water use resilience.

5. Conclusions

Our study highlights the complex interplay between soil water availability, atmospheric demand, and tree-specific factors in determining tree water use resilience to drought. We found that wetter soil and atmospheric conditions before, during and after drought generally promoted resilience, and that recurrent droughts decrease resilience. This latter finding has important implications under ongoing climate change, as most regions are experiencing or expected to experience an increase in drought frequency (IPCC, 2023). We also found that resilience could be predicted, to some extent, by species-level traits, including hydraulic and water relations traits expected to affect water use regulation by plants. However, some of the reported trait effects were inconsistent with expectations, highlighting the need for a more nuanced understanding of how trait coordination determines tree water use responses. Despite limitations in data representativity and spatial resolution, our findings provide a foundational understanding of tree water use responses to drought at relatively short timescales, emphasising the need for more detailed trait data to improve global resilience analyses.

CRedit authorship contribution statement

Brenda V. Fatecha: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jordi Martínez-Vilalta:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Maurizio Mencuccini:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Rafael Poyatos:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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